

Seonghak Lee

CONTACT INFORMATION

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EDUCATION

Chung-Ang University <i>Ph.D. candidate in Computer Science and Engineering, supervised by Prof. Junseok Kwon.</i>	Seoul, Korea Mar. 2025 – Present
Chung-Ang University <i>M.S. in Computer Science and Engineering, supervised by Prof. Junseok Kwon.</i>	Seoul, Korea Mar. 2023 – Feb. 2025
Chung-Ang University <i>B.S. in Computer Science and Engineering</i>	Seoul, Korea Mar. 2017 – Feb 2023

RESEARCH - CONFERENCE

- Jisoo Park, **Seonghak Lee**, Hyojin Park, Junseok Kwon, "SENSE: Semantic Neural Speech Synthesis from Brain Dynamics via Spatial Graph Encoding" in The Conference on Neural Information Processing Systems (**NeurIPS**), Sydney, Australia, 2026
- **Seonghak Lee**, Junhee Cho, Jisoo Park, Min-Gyu Park, Jongmin Lee, Ju Hong Yoon, Junseok Kwon, "URHead: A Unified UV-Space Representation for Joint Mesh-3DGS Optimization in Head Avatars" in Proceedings of European Conference on Computer Vision (**ECCV**), 2026
- **Seonghak Lee**^{*}, Jisoo Park^{*}, Junseok Kwon, "Lightweight Wasserstein Audio-Visual Model for Unified Speech Enhancement and Separation" in Proceedings of IEEE Automatic Speech Recognition and Understanding Workshop (**ASRU**), 2025 (*equal contribution)
- **Seonghak Lee**^{*}, Jisoo Park^{*}, Paul Hyunbin Cho^{*}, Zongkai Xia, Loucif Hebbache, "Enhancing Sidewalk Maintenance through Accurate Joint Deflection Measurement" in Proceedings of IEEE 43rd International Conference on Consumer Electronics (**ICCE**), 2025 (*equal contribution)
- **Seonghak Lee**^{*}, Jisoo Park^{*}, Junseok Kwon, "WildTalker: Talking Portrait Synthesis in the Wild" in Proceedings of European Conference on Computer Vision (**ECCV Workshop**), 3D Modeling, Reconstruction, and Generation in the Wild, 2024 (*equal contribution)
- **Seonghak Lee**^{*}, Jisoo Park^{*}, Junseok Kwon, "POTF: Prompt-based Object-centric Tensorial Field" in Proceedings of International Conference on ICT Convergence (**ICTC**), 2024 (*equal contribution), **Oral presentation**

RESEARCH - JOURNAL

- Junhee Cho, **Seonghak Lee**, Jongmin Lee, Junseok Kwon, "DiNGGA: Dirichlet and Noise-Guided Gaussian Avatar for Fine-grained Facial Detail" in **IEEE Access**, vol. 14, no. 1, pp. 118011 - 118022, 2026
- **Seonghak Lee**^{*}, Jisoo Park^{*}, Junseok Kwon, "WildTalker ∞ : Pushing the Limits of 3D Talking Portrait Synthesis in Unconstrained Environments" in IEEE Transactions on Audio, Speech and Language Processing (**T-ASLP**), vol. 34, pp. 1259–1271, 2026 (*equal contribution)
- **Seonghak Lee**, Jisoo Park, Radu Timofte, Junseok Kwon, "VT-Surf: Visual Tracking with Switching Dynamics under Time-Series Forecasting" in **IET Electronic Letters**, vol. 61, 2025
- **Seonghak Lee**, Jisoo Park, Radu Timofte, Junseok Kwon, "SOTA: Sequential Optimal Transport Approximation for Visual Tracking in Wild Scenario" in **IEEE Access**, vol. 12, no. 1, pp. 177028–177037, 2024

INDUSTRIAL EXPERIENCE

POLYGOM <i>Research Intern</i>	Gyeonggi-do, Korea Dec. 2024 – Current
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RESEARCH EXPERIENCE

DeepLearn 2026 – 13th International School on Deep Learning <i>Summer School Participant</i>	Orléans, France Jul. 2026
University of Toronto <i>Visiting Scholar</i>	Toronto, Canada Jan. 2024 – Jun 2024

AWARDS

Big Data Analytics Idea Contest by Sundo Soft (2nd Place)

Sept. 2023

- Analysis of fire service vulnerability areas and efficiency of current resource deployment (DEA) based on data such as fire, rescue, and emergency response incidents.

Big Data Analytics Idea Contest by HD Hyundai Site Solution (3rd Place)

Sept. 2023

- Award presented for proposing accident prevention technologies based on the analysis of five major accident types using CCTV footage from construction sites.

SKILLS & INTERESTS

Skills: Pytorch, Numpy, Python, Java, Git, Notion, Slack

Interests: 3D Reconstruction, Photorealistic 3D Head Avatar, Audio Enhancement and Separation